

EMAIL CAMPAIGN PLATFORM

Functional Requirements Document

Your step-by-step guide to building an in-house email platform
— contacts, templates, scheduling, tracking & analytics.

How to Use This Document

Welcome! This document tells you **exactly what to build**. Read each section before you start that part of the app. You don't need to understand everything at once.

■ Reading Order (Start Here)

1. Section 1 — Understand what you're building. 2. Section 2 — Learn the two user types (Admin & Subscriber). 3. Section 3 — See all 9 feature modules at a glance. 4. Section 4 — Dive into feature details one module at a time. 5. Section 5 — Use the screen checklist while building. 6. Section 6 — Refer to API routes when connecting frontend & backend.

Symbol	What it means
■ Must Have	Required — judges will check this.
■ Bonus	Optional extra credit — earns bonus points.
■ Watch Out	Easy to get wrong — pay attention.
■ Tip	Helpful hint to save you time.

1 What Are We Building?

You are building an **Email Campaign Platform** — a web app that lets teams create, send, and track email campaigns to large groups of people.

Think of it like building your own version of Mailchimp or Zoho Campaigns, powered by **AWS SES** (Amazon's email sending service).

What the app does — in plain English:

- Manage contacts — store email lists, import from CSV, group people into lists.
- Drag-and-drop email builder — design beautiful emails visually, no coding needed.
- Create & send campaigns — send emails to hundreds or thousands at once.
- Schedule sends — set a future date & time and the app sends automatically.
- Track opens & clicks — see who opened your email and what they clicked.
- Handle bounces & complaints — auto-remove bad email addresses.
- Manage unsubscribes — people can opt out, the app respects that always.
- Show analytics — dashboards with open rates, click rates, and more.

■ ■ The Tech Stack (Recommended)

Frontend: Any AI website builder — Lovable, Cursor, Bolt, v0, Builder.io, Webflow AI, etc. Backend: Node.js (Express) or Python (FastAPI / Flask) Database: PostgreSQL Email: AWS SES (Simple Email Service) Storage: AWS S3 (for images used in emails) Queue: AWS SQS (for sending at scale without crashing)

■ Which AI Tool Should I Use?

You can use ANY AI-powered web development tool. Popular options: • Lovable (lovable.dev) — describe your UI in plain English, get React code. • Cursor (cursor.sh) — Antigravity with AI that writes and edits your code. • Bolt (bolt.new) — full-stack app builder in the browser, no setup. • v0 by Vercel (v0.dev) — generate React components from a description. • Builder.io — drag-and-drop + AI for visual web building. Pick whichever you're most comfortable with. The goal is to ship fast!

2 User Types (Who Uses the App?)

There are **two types of users**. Each has a different experience and different permissions.

	ADMIN	REGULAR USER (Subscriber)
Who are they?	Marketing manager, IT admin, or team lead who runs the campaigns.	A person who receives emails. They don't log in.
What can they do?	Create contact lists, design email templates, send campaigns, and view reports.	Click the unsubscribe link or manage their own email preferences.
Login required?	Yes — email + password.	No — they use a unique link sent inside each email.
Main screens	Dashboard, Contacts, Templates, Campaigns, Analytics, Settings.	Unsubscribe page, Preference Centre.

Admin Roles — 3 levels inside Admin:

Role	Who	What They Can Do
Super Admin	IT / Platform Owner	Full access — AWS settings, all data, user management, billing.
Campaign Manager	Marketing team	Create lists, build email templates, send campaigns, view reports.
Viewer	Leadership / Stakeholders	Read-only — can see dashboards and reports, cannot create or edit anything.

3 App Overview — 9 Features to Build

The app has **9 feature modules**. Each is a self-contained chunk you can build one at a time. Build them in order — each one builds on the last.

#	Module Name	What It Does	Priority
M1	Login & User Management	Sign in, roles, profile settings	■ Must
M2	Contact List Management	Store contacts, import CSV, create groups	■ Must
M3	Email Template Builder	Drag-and-drop email design tool	■ Must
M4	Campaign Management	Create and manage email campaigns	■ Must
M5	Scheduling & Sending	Send now or schedule for later	■ Must
M6	Open & Click Tracking	See who opened/clicked your emails	■ Must
M7	Bounce & Complaint Handling	Auto-remove bad emails, stay AWS-compliant	■ Must
M8	Unsubscribe Management	Let people opt out safely and easily	■ Must
M9	Analytics & Reports	Dashboards, charts, campaign stats	■ Must

4 Feature Details — What to Build in Each Module

Read each module before you start building it. Every bullet marked ■ is required. ■ **Bonus** items are optional extra credit.

M1 — Login & User Management

What the Admin sees:

- A login page with email and password fields.
- After login, they land on the Dashboard.
- Profile page to update name, email, and password.
- Settings → Users page to invite teammates and assign roles.

What the system does behind the scenes:

- Stores passwords hashed (bcrypt) — never plain text.
- Issues a JWT token on login; expires after 15 min, auto-renews silently.
- Locks an IP for 15 min after 5 failed login attempts.
- Shows/hides menu items based on the user's role.

■ Tip for Freshers

Use a ready-made auth library to save time: • Supabase Auth (free, easy setup) • Clerk.dev (beautiful login UI out of the box) • NextAuth.js (if using Next.js) Don't build auth from scratch — spend that time on campaign features!

M2 — Contact List Management

This is the address book of your app. All contacts and their groups live here.

What the Admin sees:

- Contacts sidebar with two sub-pages: **Lists** and **Segments**.
- **Lists page**: grid of mailing lists — name, contact count, create/rename/delete actions.
- **List detail**: table of contacts in that list — searchable, sortable, with bulk actions.
- **Contact detail**: one person's info + which lists they're in + full email event history.
- **Import page**: upload CSV → map columns → preview → import.
- **Segments page**: build smart filters like 'opened last email but never clicked'.

Contact fields to store:

Field	Required?	Notes
Email address	■ Yes	Must be unique. Validate format before saving.
First name	Optional	Used in email personalisation — {{first_name}}
Last name	Optional	Used in email personalisation — {{last_name}}
Phone	Optional	
Status	■ Auto	active / unsubscribed / bounced / complained
Custom fields	Optional	Admin can add any extra fields: company, city, job title, etc.

Field	Required?	Notes
Source	■ Auto	How they were added: import / form / manual / API

CSV Import — step by step:

- User uploads a CSV file (maximum size: 25 MB).
- App shows a preview of the first 10 rows.
- User maps CSV column headers to contact fields using a dropdown.
- User confirms — import runs in the background.
- When done: show count of added / updated / skipped / failed rows.

■ Segment Ideas to Support

- Opened any email in the last 30 days
- Never opened any email (re-engagement target)
- Member of List A but NOT List B
- Custom field 'City' equals 'Chennai'

M3 — Email Template Builder

The most visual part of the app. Like a simple Canva — but for emails.

What the Admin sees:

- **Template Library:** grid of saved templates with thumbnails, search, category filter.
- **Editor:** two-panel layout — block palette on the left, email canvas on the right.
- **Preview toggle:** switch between desktop (600 px wide) and mobile (375 px) view.
- **HTML mode:** toggle to see and edit the raw HTML if needed.

Blocks available in the drag-and-drop palette:

Block	What it does
Header	Logo image + headline text + background colour.
Text	Rich text editor — bold, italic, links, font size, colour, alignment.
Image	Upload or paste URL, set alt text, add a clickable link, set width.
Button	CTA button — label, URL, colour, text colour, border radius, padding.
Divider	Horizontal line to separate sections — set colour and style.
Spacer	Empty vertical gap — set the height in pixels.
2-Column Layout	Two side-by-side sections, each can hold any block.
3-Column Layout	Three side-by-side sections.
Social Icons	LinkedIn, Twitter/X, Facebook, Instagram, YouTube with custom URLs.
Footer	Unsubscribe link (auto-added), mailing address, copyright text.
HTML Block	Paste any custom HTML snippet directly into the canvas.

Personalisation — merge tags:

- Any text field supports: `{{first_name}}`, `{{last_name}}`, `{{email}}`, `{{custom.company}}`
- A tag picker button opens a dropdown — no need to type the tag manually.
- Preview mode replaces tags with sample data from the first contact in the target list.

Template rules:

- Generated HTML must use **inline CSS only** — email clients strip external stylesheets.
- Must look correct on both desktop (600 px) and mobile (375 px).
- Tracking pixel and unsubscribe link are added automatically at send time.
- Ship at least **5 starter templates**: Welcome, Newsletter, Promotional, Event Invite, Training Notice.

■ **Tip: Don't Build the Editor from Scratch**

Use an open-source email editor library: • Unlayer (unlayer.com) — free embeddable drag-and-drop editor • MJML (mjml.io) — responsive email framework, auto-generates inline CSS • GrapeJS (grapesjs.com) — open-source visual builder
Embedding one of these saves weeks of work!

M4 — Campaign Management

A Campaign = one email send event. Pick a template, choose recipients, hit send.

Campaign creation — 4-step wizard:

Step	What the user fills in
Step 1 Details	Campaign name (internal only), email Subject line, Preview text (shown in inbox before opening), Sender name, Sender email address, Reply-to email (optional).
Step 2 Recipients	Pick one or more contact lists or saved segments. Optionally exclude specific lists. App shows estimated recipient count after removing duplicates and unsubscribed contacts.
Step 3 Design	Pick an existing template from the library — OR open the editor to create a new one inline. Both paths land on the same editor.
Step 4 Review	Full rendered email preview. Send a test to up to 5 email addresses. Then choose: Send Now OR Schedule (pick date, time, timezone). Shows estimated recipient count before confirming.

Campaign list page:

- Table showing: name, status badge, recipient count, send date, open rate, click rate.
- Status badges: Draft / Scheduled / Sending / Sent / Paused / Cancelled.
- Actions per campaign: Edit (draft only), Duplicate, View Report, Delete.
- Filters: by status, date range, and tags.

Other campaign features:

- **Test Send**: send a preview to up to 5 addresses before the real send. Not counted in analytics.
- **Duplicate**: copy all settings into a new draft campaign.
- ■ **Bonus — Re-send**: re-send to contacts who did NOT open the original email.

M5 — Scheduling & Sending

Send Now:

- Clicking Send Now queues all recipient emails into AWS SQS immediately.
- A background worker reads from the queue and calls AWS SES for each contact.
- Must respect AWS SES rate limits — don't send too fast or SES throttles you.

Scheduled Send:

- User picks a future date, time, and timezone.

- App triggers the send at the right UTC time (use AWS EventBridge or a cron job).
- User can edit the schedule up to 15 minutes before dispatch.
- User can cancel at any time before the send starts.

Send Progress:

- While sending, show a live progress bar: '12,450 of 50,000 sent'.
- Update the count every 5 seconds using polling or WebSocket.
- Pause button stops the queue. Resume continues from where it left off.

■ ■ AWS SES Sending Limits — Read This!

New SES accounts start in sandbox mode — you can only send to verified email addresses, with a 200 email/day limit. For the demo, this is fine — just verify a few test addresses. For production: request a sending limit increase via AWS Console → SES → Account Dashboard → Request Production Access.

- ■ Bonus: Recurring campaigns — daily, weekly, or monthly repeating sends.

M6 — Open & Click Tracking

How Open Tracking works — step by step:

- When sending an email, the app adds an invisible 1×1 pixel image at the bottom.
- The image URL points to your backend: **/track/open?uid=ABC123**
- When the email client loads the image, your server records an 'opened' event.
- The uid is a unique code that identifies the contact + campaign combination.

How Click Tracking works:

- Every link in the email is rewritten to pass through your server first.
- Original: **https://example.com/offer**
- Rewritten: **/track/click?uid=ABC123&url=https://example.com/offer**
- Your server records the click, then instantly redirects to the original URL (HTTP 302).

What to record for each event:

- Which contact (contact_id), which campaign (campaign_id), when (timestamp).
- For clicks: which specific link was clicked (original URL).
- Device info: IP address and browser/email client (from User-Agent header).

■ Use AWS SES Native Tracking

AWS SES Configuration Sets can automatically track opens and clicks for you — no custom pixel needed. Set up a Configuration Set in SES, point events to an SNS topic, and your backend receives all events via webhook. This is the recommended approach.

M7 — Bounce & Complaint Handling

■ ■ AWS SES Rule — This is Non-Negotiable

AWS requires: • Bounce rate must stay below 2%. • Complaint rate must stay below 0.1%. If you exceed these thresholds, AWS will pause your sending account. Your app **MUST** automatically stop sending to bounced and complained addresses.

Types of email problems & how to handle them:

Type	What it means	What your app must do
Hard Bounce	Email address doesn't exist or domain is invalid (permanent).	Immediately add to suppression list. Set contact status = bounced. Never send again.
Soft Bounce	Temporary issue — inbox full, mail server down.	Count soft bounces. After 3 in 30 days → treat as hard bounce.
Complaint	Person marked your email as spam.	Immediately add to suppression list. Set contact status = complained. Never send again.

How your app receives these notifications:

- AWS SES → sends bounce/complaint events to an SNS topic.
- SNS → forwards messages to an SQS queue.
- Your backend → has a worker that reads from SQS and processes each message.
- The worker updates the contact status and adds them to the suppression list.

Suppression List page (Admin):

- Table of all suppressed emails: address, reason (bounced/complained), date added.
- Admin can manually add or remove entries (with reason logged in audit trail).
- CSV import automatically skips any email found in the suppression list.

M8 — Unsubscribe Management

People must be able to opt out easily. This is both a legal requirement and best practice.

Every email must include an unsubscribe link:

- Added automatically in the email footer — the admin doesn't need to remember.
- Link URL: `/unsubscribe?uid=ABC123` (uid identifies the contact + campaign).
- Clicking the link instantly marks the contact as unsubscribed.

Unsubscribe confirmation page (public — no login needed):

- Shows a friendly message: 'You have been unsubscribed from [List Name].'
- Option to re-subscribe if they changed their mind.
- Branded with the organisation's name and logo.

One-Click Unsubscribe — required by Gmail & Yahoo (2024+):

- Every email must include: **List-Unsubscribe** and **List-Unsubscribe-Post** headers.
- Your backend handles a POST request to the unsubscribe URL immediately.
- This lets Gmail/Apple Mail show a built-in 'Unsubscribe' button at the top of the email.
- ■ Bonus: Preference Centre — instead of a full unsubscribe, let people pick which lists to stay on.

M9 — Analytics & Reports

Main Dashboard widgets (shown when Admin logs in):

Widget	What it shows
Total Contacts	Total count with a growth trend (last 30 days).
Emails Sent	Total sent in the selected date range.

Widget	What it shows
Avg. Open Rate	% of emails that were opened. Industry average: ~21%.
Avg. Click Rate	% of emails where a link was clicked. Average: ~2.5%.
Bounce Rate	Turns amber/red if above 2% (AWS danger zone).
Complaint Rate	Turns amber/red if above 0.1%.
Top 5 Campaigns	Table of best-performing campaigns by open rate.
Recent Activity	Live feed of the last 20 tracking events.

Campaign Report page (one per sent campaign):

- Summary numbers: Sent, Delivered, Unique Opens, Total Opens, Unique Clicks, Total Clicks, Bounces, Complaints, Unsubscribes.
- **Open rate over time:** line chart — hourly for first 48h, then daily.
- **Link click breakdown:** table of every tracked link with click count and %.
- **Device breakdown:** desktop vs mobile vs tablet (from User-Agent).
- **Per-contact activity:** each recipient's open/click/bounce/unsubscribe status.

Key metrics formula sheet:

Metric	Formula	Target
Open Rate	$\text{Unique Opens} \div \text{Delivered} \times 100$	Above 20%
Click Rate	$\text{Unique Clicks} \div \text{Delivered} \times 100$	Above 2%
Bounce Rate	$\text{Bounces} \div \text{Sent} \times 100$	Keep below 2%
Unsubscribe Rate	$\text{Unsubscribes} \div \text{Delivered} \times 100$	Keep below 0.5%

Export:

- Campaign report → exportable as **CSV** (all per-contact event data).
- Contact list → exportable as **CSV** with all fields.
- ■ Bonus: PDF export of campaign summary report.

5 Screens to Build — Complete Checklist

Every screen your app needs. Use this as a build checklist. All ■ screens are required.

Admin Screens (requires login):

#	Screen Name	URL Route	Key Elements on This Screen
1	Login	/login	Email + password form, error messages, forgot password link.
2	Dashboard	/	KPI cards, open/click rate charts, recent activity feed.
3	Contact Lists	/contacts/lists	Grid of all lists, create new list button, list stats.
4	Contact List Detail	/contacts/lists/:id	Contact table, search bar, filter, bulk actions, import button.
5	Contact Detail	/contacts/:id	Profile fields, event history timeline, list memberships.
6	Import Contacts	/contacts/import	File upload widget, column mapper, 10-row preview, progress bar.
7	Segments	/contacts/segments	Segment list, rule builder with AND/OR logic, live count.
8	Template Library	/templates	Thumbnail grid, search, category filter, 5 starter templates.
9	Template Editor	/templates/:id/edit	DnD canvas, block palette, preview toggle (desktop/mobile), HTML tab.
10	Campaigns List	/campaigns	Table with status badges, date, open rate, click rate, actions.
11	New Campaign — Step 1	/campaigns/new/details	Name, subject, preview text, from name, from email fields.
12	New Campaign — Step 2	/campaigns/new/recipient s	List/segment multi-select, exclusion picker, recipient count.
13	New Campaign — Step 3	/campaigns/new/design	Template selector grid + button to open editor inline.
14	New Campaign — Step 4	/campaigns/new/review	Full preview, test send button, schedule date/time/timezone picker.
15	Campaign Report	/campaigns/:id/report	All stats, line chart, link table, device breakdown, contact table.
16	Suppression List	/settings/suppression	Table of bounced/complained emails, add/remove manually.
17	Organisation Settings	/settings/org	Name, logo upload, default from-email, AWS SES config fields.
18	User Management	/settings/users	User list, invite by email, change roles, deactivate user.

Public Screens (no login — for email recipients):

#	Screen Name	URL Route	Key Elements on This Screen
19	Unsubscribe Page	/unsubscribe	Friendly confirmation, re-subscribe option, organisation branding.
20	Preference Centre	/preferences	■ Bonus — per-list subscription toggles for the subscriber.

6 Backend API Routes

All routes except public tracking and unsubscribe routes require a JWT token in the **Authorization: Bearer <token>** header.

Auth

Method	Path	What it does
POST	/api/auth/login	Login → returns JWT access token + refresh token.
POST	/api/auth/logout	Revoke the current session.
GET	/api/auth/me	Return the logged-in user's profile.

Contacts & Lists

Method	Path	What it does
GET	/api/lists	Get all contact lists.
POST	/api/lists	Create a new list.
GET	/api/lists/:id	Get one list + stats (count, bounce %, open %).
PUT	/api/lists/:id	Rename or update a list.
DELETE	/api/lists/:id	Delete a list.
GET	/api/lists/:id/contacts	Get contacts in a list (paginated + searchable).
POST	/api/lists/:id/import	Upload CSV file and start async import job.
GET	/api/contacts/:id	Get one contact + full email event history.
POST	/api/contacts	Add a single contact manually.
PUT	/api/contacts/:id	Update a contact's fields.
DELETE	/api/contacts/:id	Delete a contact.
GET	/api/segments	List all saved segments.
POST	/api/segments	Create a segment with filter rules.
GET	/api/segments/:id/count	Live contact count matching a segment's rules.

Templates

Method	Path	What it does
GET	/api/templates	List all templates.
POST	/api/templates	Create a template (saves JSON blocks + HTML).
GET	/api/templates/:id	Get one template with its blocks and HTML.
PUT	/api/templates/:id	Save/update a template.
DELETE	/api/templates/:id	Delete a template.

Method	Path	What it does
POST	/api/templates/:id/duplicate	Copy a template into a new draft.
POST	/api/templates/upload-asset	Upload an image to S3, get back a CDN URL.

Campaigns

Method	Path	What it does
GET	/api/campaigns	List all campaigns (filter by status, date, tag).
POST	/api/campaigns	Create a new draft campaign.
GET	/api/campaigns/:id	Get campaign details and settings.
PUT	/api/campaigns/:id	Update a draft campaign.
DELETE	/api/campaigns/:id	Delete a draft campaign.
POST	/api/campaigns/:id/send	Send the campaign immediately.
POST	/api/campaigns/:id/schedule	Schedule the campaign for a future date/time.
POST	/api/campaigns/:id/pause	Pause a currently sending campaign.
POST	/api/campaigns/:id/resume	Resume a paused campaign from where it stopped.
POST	/api/campaigns/:id/cancel	Cancel a scheduled campaign.
POST	/api/campaigns/:id/test	Send test email to up to 5 addresses.
GET	/api/campaigns/:id/report	Get full analytics report for this campaign.
GET	/api/campaigns/:id/progress	Get live sending progress (sent / total).

Tracking & Public Routes (no token needed)

Method	Path	What it does
GET	/track/open	Record email open event. Return a 1×1 transparent PNG image.
GET	/track/click	Record link click. Redirect to original URL (HTTP 302).
GET	/unsubscribe	Show the unsubscribe confirmation page.
POST	/unsubscribe	Process unsubscribe (used by List-Unsubscribe-Post header).
GET	/preferences	■ Show preference centre page.
POST	/preferences	■ Update the subscriber's list preferences.
POST	/webhooks/ses	Receive bounce/complaint/delivery events from AWS SNS.

7 Database — What Tables to Create

Use **PostgreSQL**. Every table you need is listed below. Refer to this when setting up your database or ORM models.

Table	Key Fields	What it stores
users	id, email, password_hash, role, org_id, created_at	Admin users of the platform.
organisations	id, name, logo_url, from_email, ses_config_set, aws_region	Your organisation's global settings.
contact_lists	id, org_id, name, description, tags[]	Named mailing lists (e.g. 'Newsletter', 'Training').
contacts	id, org_id, email (unique), first_name, last_name, status, custom_fields (JSON), source	Individual email subscribers.
contact_list_members	contact_id, list_id, subscribed_at	Which contacts belong to which lists (many-to-many).
segments	id, org_id, name, rules (JSON)	Saved dynamic filter rules.
templates	id, org_id, name, category, blocks (JSON), html, thumbnail_url	Email templates with block structure and rendered HTML.
campaigns	id, org_id, name, subject, preview_text, from_name, from_email, status, template_id, scheduled_at, timezone	All campaigns and their settings.
campaign_sends	id, campaign_id, contact_id, ses_message_id, status, sent_at	One record per recipient per campaign.
email_events	id, contact_id, campaign_id, event_type, occurred_at, ip, user_agent, metadata (JSON)	All tracking events: opens, clicks, bounces, complaints, unsubscribes.
suppression_list	id, org_id, email, reason, suppressed_at	Emails that must NEVER receive any campaign.
import_jobs	id, list_id, status, total_rows, added, updated, skipped, errored, started_at	Tracks progress of CSV import jobs.

■ event_type values for email_events table

sent | delivered | opened | clicked | bounced | complained | unsubscribed

8 AWS SES Setup — Step by Step

AWS SES is the engine that actually sends the emails. Here's how to set it up from scratch — even if you've never used AWS before.

Step 1 — Create an AWS account

- Go to **aws.amazon.com** → sign up or log in.
- Search for **SES** (Simple Email Service) in the top search bar.

Step 2 — Verify your email or domain

- SES → Verified Identities → Create Identity.
- **For the (easiest):** verify a single email address. AWS sends a confirmation email — click the link.
- **For production:** verify your domain and set up DKIM (SES guides you through each step).

Step 3 — Create a Configuration Set

- SES → Configuration Sets → Create → name it **email-platform-events**.
- Add an event destination: choose **SNS topic**.
- Select all event types: Send, Delivery, Open, Click, Bounce, Complaint.

Step 4 — Connect SNS → SQS (for bounce/complaint processing)

- AWS SNS → Create a new topic.
- AWS SQS → Create a new queue → Subscribe it to your SNS topic.
- Your backend worker reads from this SQS queue to handle bounces and complaints.

Step 5 — Request production access

- By default SES is in 'sandbox' — you can only send to verified addresses (fine for hackathon).
- For real users: SES → Account Dashboard → Request Production Access.

Step 6 — Install the AWS SDK in your backend

Language	Install Command
Node.js	<code>npm install @aws-sdk/client-ses @aws-sdk/client-s3 @aws-sdk/client-sqs</code>
Python	<code>pip install boto3</code>

■ ■ Never Hardcode AWS Keys in Your Code

Store credentials as environment variables: `AWS_ACCESS_KEY_ID=your_key`
`AWS_SECRET_ACCESS_KEY=your_secret` `AWS_REGION=ap-south-1` Use a `.env` file locally and set environment variables on your hosting platform (Render, Railway, Vercel, etc.). Never commit your `.env` file to GitHub — add it to `.gitignore`!

9 What to Submit — Deliverables & Judging

Required deliverables:

#	Deliverable	Details
1	GitHub Repo	Public repository with all source code — frontend + backend.
2	README.md	How to run locally. List all environment variables. Include AWS SES setup steps. Add an architecture diagram.
3	Live Demo URL	Fully deployed and working. Use Vercel/Netlify for frontend, Render/Railway/Fly.io for backend.
4	Demo Video	5-minute screen recording walking through all 9 modules working end-to-end.
5	API Docs	Postman collection OR OpenAPI/Swagger spec covering all endpoints.

The Deadline for Submitting the Web Application is 12th May.

Please Send the GitHub Link or Application URL to: Mail - programassociate@theaischool.co

You're all set — start building! ■

Start with **M1 (Login)** to get auth working,
then **M2 (Contacts)** to build the address book,
then the **Template Builder** and **Campaigns**.

Use any AI tool — Lovable, Cursor, Bolt, v0, or anything else you're comfortable with.
This document is your map. Build one section at a time. Good luck!